

Step-1: Efficient Tuple Condensation for LLM-Based Entity Resolution

1 Motivation

In Large Language Model (LLM) based Entity Resolution (ER), the predominant bottleneck is the **cost associated with input prompt tokens**. A substantial fraction of these tokens stems from inefficient tuple representations. Standard approaches often serialize numerous attributes—including those that are missing, redundant, noisy, or excessively long—and redundantly process them for every pairwise comparison.

Such verbosity exacerbates token consumption and inference latency, while potentially compromising accuracy by overwhelming the model with irrelevant context.

1.1 Illustrative Example

Consider a standard movie database record. A naive serialization includes all available fields, regardless of utility:

Verbose Input:

[Title: Dune] [Year: 2021] [Director: Denis Villeneuve] [Synopsis: A mythic and emotionally charged hero's journey, "Dune" tells the story of Paul Atreides... (200+ tokens)] [Format: 4K UHD] [User Rating: null]

To overcome this, we can propose creating a **compact, identity-focused description** for each record. This targeted representation preserves the critical evidence necessary to determine entity equivalence while systematically eliminating extraneous text:

Condensed Input:

[Title: Dune] [Year: 2021] [Director: Denis Villeneuve]

By distilling the tuple down to core identity attributes, we significantly reduce the prompt length per pair while focusing the LLM strictly on the signals that dictate a match.

2 The Condensation Strategies

Intuitively, we can consider the following condensation strategies to make tuple representations shorter and more efficient. The main goal is simple: use fewer tokens while keeping matching

accuracy high. Think of it as finding the best balance between speed/cost and quality.

Absolutely. Focusing on the two strategies that deliver the most impact is a smart and practical choice. Here is the revised, streamlined section:

1. Attribute Pruning (Choosing Which Attributes to Keep): Keep only the attributes that really help tell entities apart in your dataset.

- *Keep:* Important fields like brand, model, and key words (for products); or title, authors, year, and DOI (for papers).
- *Remove:* Fields that are often empty, always the same, or not useful for matching (e.g., seller comments, shipping details, or promotional text).
- *Example (Product Record):*
 - **Before (Verbose):** *[Title: Wireless Headphones] [Brand: Sony] [Model: WH-1000XM4] [Color: Black] [Price: \$279.99] [Seller: "TechDeals123"] [Shipping: "Free 2-day shipping"] [Promo: "Limited time offer!"] [Description: "Experience premium sound quality... (150+ words)"]*
 - **After (Pruned):** *[Title: Wireless Headphones] [Brand: Sony] [Model: WH-1000XM4] [Color: Black]*
 - *Why:* Brand, model, and title help distinguish this product from others. Seller notes and promo text rarely help decide if two records match the same item. **Impact: High token savings.**

2. Value Compression (Shortening Long Fields): For fields that contain useful but very long text (like product descriptions), replace the full text with a shorter version.

- *How:* Keep the most important words—like key terms, unique IDs, and numbers with units—and remove filler or repetitive language.
- *Example (Product Description):*
 - **Before (Full Text):** *"The Sony WH-1000XM4 wireless headphones deliver exceptional sound quality with industry-leading noise cancellation technology. Featuring a sleek, comfortable design with plush ear cushions, these headphones are perfect for long listening sessions. Battery life extends up to 30 hours on a single charge, with quick charging capability providing 5 hours of playback from just 10 minutes of charging. Compatible with Bluetooth 5.0, NFC, and voice assistants..."*
 - **After (Compressed):** *"Sony WH-1000XM4: noise cancellation, Bluetooth 5.0, 30h battery, quick charge, NFC"*
 - *Why:* The compressed version keeps the unique specs that help identify the product while removing marketing language that adds tokens but little matching value. **Impact: High token savings.**

3 Implementation Pathways

For this project, you are encouraged to implement the above condensation strategies through the following methods:

3.1 Manual Inspection

Difficulty: Low — Data Needed: None

Idea: Manually define a fixed subset of attributes to retain across all records.

How: Analyze the dataset schema to filter out visibly verbose or low-signal fields (e.g., *description*, *seller_notes*, *shipping*), while preserving core identity markers (e.g., *title*, *brand*, *model*).

Best For: Small, structurally homogeneous datasets (where records follow a highly consistent format) or scenarios lacking labeled data.

Risk: May inadvertently discard attributes that provide critical disambiguating signals in edge cases.

3.2 Data-Driven Heuristics (Statistics & Regex)

Difficulty: Medium — Data Needed: Unlabeled Dataset

Idea: Leverage corpus-level statistics to inform pruning decisions, alongside lightweight, pattern-based rules for field compression.

How (Pruning): Calculate field-level missingness (the proportion of null values) and uniqueness (value diversity). Discard attributes that are predominantly empty or exhibit near-constant values across the dataset.

How (Compression): Deploy regular expressions (Regex) to extract high-signal tokens—such as numeric specifications, standard identifiers, or capitalized keywords—from lengthy textual descriptions, systematically stripping away boilerplate language.

Best For: Large-scale datasets where manual schema inspection is impractical.

Risk: Relying purely on broad statistical signals risks overlooking semantically vital, albeit infrequently populated, attributes.

3.3 Supervised or LLM-Guided Selection

Difficulty: High — Data Needed: Labels or LLM Access

Idea: Deduce attribute importance directly from labeled matching evidence or the internal reasoning signals of an LLM.

How (Supervised): Using labeled match/non-match pairs, train a lightweight classifier (e.g., Logistic Regression) on field-level similarity features. Analyzing the learned weights helps isolate the most discriminative fields.

How (LLM-Guided): Prompt an LLM using a representative sample of record pairs (e.g., “*Which fields were most influential in deciding whether these two records match?*”). Aggregating these responses yields a prioritized list of attributes to retain.

Best For: Highly heterogeneous or semantically complex datasets where attribute importance fluctuates significantly across records.

Risk: Incurs a higher initial setup cost and latency due to the necessity of acquiring labeled data or executing preliminary LLM queries.

4 Task Roadmap

To guide the execution of this project, we propose a three-week roadmap. This structured approach aims to systematically evaluate the condensation strategies against established baselines:

- 1. Proposed Timeline (3 Weeks):** We suggest dedicating the first week to implementing the foundational methods (Manual Inspection and Data-Driven Heuristics) and establishing the baselines. The remain weeks can then focus on exploring the more complex, resource-intensive Supervised or LLM-Guided Selection pathway.
- 2. Datasets:** We recommend utilizing three standard entity resolution benchmark datasets that you have previously explored. Ideally, these should represent different domain challenges (e.g., variations in schema homogeneity or text verbosity) to test the robustness of the condensation strategies.
- 3. Baselines & Models:** For a baseline comparison, please utilize a standard pairwise matching prompt. To manage scale and computational overhead, we suggest applying the simple blocking strategies outlined in DeepER prior to the LLM matching step. A proposed baseline prompt is as follows:

System Instruction:

You are an expert in data integration and entity resolution. Your task is to determine whether the following two records refer to the exact same real-world entity.

Record A: {Record_A}

Record B: {Record_B}

Question: *Do these records match? Please answer strictly with 'Yes' or 'No', followed by a brief, one-sentence justification.*

We encourage calling established, highly efficient models for this task, such as GPT-4.1-mini and Gemini 3 Flash.

- 4. Evaluation Metrics:** To comprehensively assess the trade-offs between matching quality and computational efficiency, please track the following metrics:

- *Accuracy:* F1-score, Recall, and Precision.

- *Efficiency:* Total input token usage, estimated monetary cost, and end-to-end inference time.

We suggest summarizing these results in a clear, comparative tabular or visual format, accompanied by a thoughtful analysis of your findings.

5. **Open Exploration:** As this is an iterative and exploratory process, please feel free to propose, experiment with, or integrate your own condensation heuristics or ideas beyond the suggested pathways.